



OPEN Traffic sign detection method based on improved YOLOv8

Gaihua Wang^{1,2✉}, Peng Jin^{2,3}, Zhiwei Qi⁴ & Xiaohuan Li^{2,3}

Traffic sign detection is crucial in intelligent transportation and assisted driving, providing favourable support for driving safety and prevention of traffic accidents. Aiming at the current traffic sign detection problems of leakage, misdetection and low detection accuracy of small targets, a traffic sign detection method based on improved YOLOv8n is proposed. Firstly, the Neck part of YOLOv8 is improved by designing a module that combines Attention Scale Sequence Fusion with the P2 small target detection layer (AFP) to enhance the feature extraction capability of the YOLOv8 network, enabling it to capture more small target features. Secondly, a lightweight convolution module, LWConv, is designed, based on which the Bottleneck structure of Cross-convolution with two filters (C2f) in YOLOv8 is reconstructed and named LW_C2f, effectively reducing the model size and parameters. Finally, the loss function of the original YOLOv8 is replaced with the Wise-IoU loss function, which improves the network's bounding box regression performance and reduces the negative impact of low-quality samples. The experimental results show that the mean average precision (mAP50) of the improved model on the TT100K dataset is increased by 5.7% compared to the YOLOv8 model, while the number of parameters and the model size are reduced by 0.6 M and 0.8 MB, respectively.

Keywords YOLOv8, Lightweight, Traffic sign detection, Small target detection layer, Wise-IoU

With the development of urbanisation, the number of motor vehicles increases year by year, and this trend leads to a more complex traffic environment. Traffic sign detection and recognition, as an important part of Intelligent Transport System(ITS), and achieving efficient and highly accurate traffic sign detection is crucial for applications such as driverless and assisted driving. The technology recognises traffic signs such as speed limit signs, prohibition signs, warning signs and other traffic signs in images or videos, providing the necessary information to drivers and self-driving cars. It ensures the smooth operation of the road network, effectively reduces or prevents traffic accidents, and ensures road travelling safety to a certain extent. However, current traffic sign detection techniques still face challenges in detecting small target objects. For example, target detection algorithms do not work well when dealing with complex scenes such as occlusion, illumination changes, and target scale changes^{1,2}. In addition, the robustness and versatility of the algorithm need to be further enhanced to ensure effective operation in different environments and tasks^{3,4}. Therefore, in order to further investigate and validate the adaptability and performance of existing methods in complex traffic scenarios, it is necessary to conduct experimental evaluations under conditions that are closer to the real environment.

Based on the above considerations, we use the TT100K⁵ dataset in our experiments. This dataset is collected from street-view images across multiple cities in China and includes a wide range of traffic sign categories and complex environmental conditions such as occlusion, lighting changes, and scale variations. Its diversity and real-world complexity make it an ideal benchmark to evaluate the practical effectiveness of detection algorithms and highlight the real-world relevance of this study.

To cope with the above challenges, many scholars have proposed a variety of innovative target detection algorithms in recent years. For example, Literature 1⁶ uses a cascaded Region-based Convolutional Neural Network (R-CNN) to obtain multi-scale pyramid features. By jointly training and fusing the output bounding boxes from previous stages, it significantly improves detection accuracy; however, this approach is not suitable for real-time applications. Literature 2⁷ demonstrates that the Detection Transformer (DETR) model can effectively capture the contextual information within an image, but it demands substantially higher computational resources. Literature 3⁸ designed cluster multi-scale attention pyramid network and adaptive pyramid convolution to solve the problem of small traffic sign detection, but it is not effective in complex scenes and requires multiple debugging. Literature 4⁹ proposed a feature difference model for evaluating the effect of

¹School of Digital Arts, Xi'an University of Posts and Telecommunications, Xi'an 710100, China. ²Shaanxi Provincial Key Laboratory of Intelligent Media, Xi'an 710100, China. ³Computer Institute, Xi'an University of Posts and Telecommunications, Xi'an 710100, China. ⁴School of Education, Yunnan University, Kunming 650091, China. ✉email: ghw@xupt.edu.cn

convolution operations on features and achieved good results in traffic sign detection, however, due to the small size of the dataset, the experimental results may be affected by randomness. Literature ^{5,10} applies Top-hat and Bottom-hat transformations with morphological operations to enhance the contrast between moving objects and the background, improving detection robustness. However, its reliance on morphological processing and feature point extraction makes it sensitive to video quality and lighting variations, leading to reduced performance in complex or occluded scenes.

In addition, You Only Look Once (YOLO) remains a fundamental approach in the field of object detection. Among its many versions, the YOLOv5 model achieves a good balance between detection performance and deployment flexibility through a modular design and adaptive anchor box mechanism. However, it offers limited improvements in feature representation and lacks generalization capability for non-standard targets. The YOLOv7 model significantly enhances detection accuracy by incorporating an efficient layer aggregation network and a dynamic label assignment strategy, but its multi-branch architecture increases deployment complexity. YOLOv8 further improves feature representation and small object detection performance by introducing an anchor-free decoupled detection head, an enhanced feature fusion structure, and a unified multi-task learning framework, thereby addressing the limitations of YOLOv5 in complex scenarios. Compared to YOLOv7, YOLOv8 simplifies the network architecture, significantly reducing deployment complexity and inference overhead while maintaining high detection accuracy. Nevertheless, although YOLOv8 achieves better performance in small object detection than its predecessors, it still suffers from missed detections and false positives under complex conditions such as occlusion and dense object distributions, which affect overall detection accuracy. Additionally, the use of traditional convolution operations leads to a large number of parameters and high computational cost. Moreover, YOLOv8's loss function places excessive emphasis on the regression of low-quality samples^{11,12}, lacking a dynamic non-monotonic focusing mechanism and an intelligent gradient gain allocation strategy¹³. The dynamic non-monotonic focusing mechanism evaluates anchor quality via an outlier degree: high-quality anchors (low outlier degree) and low-quality anchors (high outlier degree) both receive lower gradient gains, reducing harmful gradients and competition. This allows the model to focus on optimizing medium-quality anchors, thereby enhancing overall regression performance.

In order to cope with these challenges, as well as to solve the current problems of YOLOv8, this paper uses the YOLOv8 algorithm as a baseline model and improves on it. The Neck part of YOLOv8 is improved by designing AFP module to achieve effective detection of small targets. Meanwhile, the lightweight LW_C2f module is designed to replace part of the C2f module of YOLOv8, which can reduce the number of parameters and computation of the model. Finally, the Wise-IoU loss function is used to replace the original loss function of YOLOv8, which combines the dynamic non-monotonic focusing mechanism with the wise gradient gain assignment strategy to emphasise the average-quality samples in the data and reduce the negative impact of low-quality samples.

The main contributions of this paper are as follows:

- Firstly, the AFP module is designed, including the introduction of the Attention Scale Sequence Fusion based You Only Look Once framework (ASF)¹⁴ and incorporating the P2 small target detection layer. This module is used to improve the Neck part of YOLOv8 to achieve effective detection of small target regions by highlighting key features and suppressing invalid features.
- Secondly, we designed a lightweight convolution LWConv to reconstruct part of the C2f of YOLOv8, named LW_C2f module. While ensuring accuracy, it reduces the number of parameters and model size, which is conducive to deployment on mobile devices.
- Lastly, we replace the original CIoU loss function with Wise-IoU¹⁵ to optimise the training model, incorporating a dynamic non-monotonic focusing mechanism to make the object bounding box regression more accurate and reduce the negative impact of low-quality samples.

Related works

Traditional traffic sign detection methods

In the past few decades, a large number of studies have been conducted based on color or shape detection methods, each of which proposed different algorithms to solve the current problem. Color-based traffic sign detection methods mainly rely on the color segmentation of the image, and then detect and extract different regions. Kiran et al.¹⁶ used color information to detect and classify the shape of traffic signs from image sequences. The hue and saturation components were enhanced by using a lookup table, and better shape classification performance was achieved by using the “distance to the boundary” and “distance to the center” features of the segmented spots. Yang et al.¹⁷ proposed a color probability model to convert the input image into a probability map, extract traffic signs by finding the most stable extreme regions on these maps, and adopt an integral channel feature detector to eliminate false positives of the proposals. Although color-based methods improve real-time operation and accuracy, the detection results may be easily disturbed due to challenging conditions such as illumination intensity and target scale changes. In addition, in the case of faded or damaged traffic signs, the detection performance of the algorithm may be severely affected, resulting in missed detections and false detections.

Shape-based traffic sign detection methods start with noise reduction through preprocessing, followed by extraction of shape features of the image through edge detection and shape detection, and finally detection using a classifier. Boumediene et al.¹⁸ proposed an effective algorithm for detecting triangular traffic signs based on the RSLD algorithm on grey scale images. The algorithm converts the detection of triangles into the detection of simple line segments with low computational effort. Yıldız et al.¹⁹ improved the detection performance by a shape-based approach that takes triangles, circles and squares into account, identifies the shapes by connecting the components, and removes shapes without traffic signs from the image. Shape-based detection methods can

reduce interference from color fading but may perform poorly when traffic signs are deformed or viewed from different angles.

Deep learning based detection methods

Unlike traditional traffic sign detection algorithms, deep learning algorithms are trained with a large amount of data to automatically learn image features and complete feature extraction to achieve traffic sign detection. There are two main deep learning-based detection methods in common use: the R-CNN^{20–22} series of algorithms based on two-stage target detection and the YOLO series of algorithms based on single-stage target detection^{23–26}.

Lee et al.²⁷ proposed a convolutional neural network (CNN) algorithm built upon the Single Shot MultiBox Detector (SSD) framework to simultaneously estimate traffic sign locations and boundaries. They formulated the boundary estimation task as a two-dimensional pose and shape classification problem, which was efficiently solved by the CNN as a pose and shape prediction task. Wang et al.²⁸ employed Res2Net: A Multi-scale Backbone Architecture for Image Classification as the backbone of the Faster R-CNN model and utilized the Region Proposal Network (RPN) to extract candidate boxes, thereby optimizing the network and enabling the acquisition of more discriminative image features. However, traffic signs in real situations are not detected under perfect ideal conditions and the presence of small targets in traffic signs is often difficult to detect, so the current problem of low detection of small targets in traffic sign images under natural conditions needs to be addressed urgently. Also unlike traffic signs that are directly observed by a human, machines often have difficulty in recognising traffic signs that are far away or have blurred images very quickly, hence many detection methods have been developed to address this problem.

Mohd et al.²⁹ and Jiang et al.³⁰ both enhanced the network structure of YOLOv3 by making a series of improvements to enhance the network's effectiveness in detecting images of small targets. Gomaa et al.³¹ proposed an improved version combining detection-based CNN with YOLOv4 technique and background subtraction, which effectively reduces the data preparation cost but lacks a corresponding error correction mechanism. Wang et al.³² improved the detection performance of the YOLOv5 network for multi-scale targets by proposing an improved feature pyramid model with an adaptive attention module (AAM) and a feature enhancement module (FEM) to reduce the loss of information during the feature map generation process and to enhance the representation of the feature pyramid. Li et al.³³ improved the detection performance of the YOLOv7 network for multi-scale targets by integrating a hybrid module of self-attention and convolution into the added small target detection layer to capture additional feature information to improve the accuracy of traffic sign detection, while also increasing the amount of computation and operations. Zhang et al.³⁴ solved the problem of insufficient feature fusion due to too small targets by designing a dual-path feature fusion network for YOLOv8, but the model is not very generalisable.

Methods

In order to solve the problems of leakage, misdetection and low accuracy of small target detection in traffic sign detection, and considering the demand of model lightweighting, this paper takes YOLOv8n model as the baseline model and improves it. Firstly, the ASF module and the small target detection layer are introduced in the Neck part of YOLOv8 to fuse features from different scale feature maps and to increase the extraction capability of small targets. Secondly, a lightweight convolution module is designed, based on which the Bottleneck structure is reconstructed and finally integrated into a new module LW_C2f, thus replacing part of the C2f module in YOLOv8. This replacement improves the computational efficiency, optimises the memory access, and reduces the number of parameters of the model while ensuring the accuracy. Finally, in order to improve the convergence ability of the classification loss of the network, the loss function is optimised with Wise-IoU to make the model more stable and easy to converge. The network structure of the improved model is shown in Fig. 1.

It is worth stating that, as can be seen in the network structure diagram, the ASF is divided into ASF1 and ASF2 due to some differences in their functionality. The ASF1 module is to integrate feature maps from different scales in order to merge multi-scale information to enhance the model's performance. Specifically, the input contains feature maps at three scales (low scale *l*, medium scale *m* and high-scale *s*), and each scale represents different spatial resolution information. In the forward propagation process, adaptive maximum pooling and adaptive average pooling operations are first performed on the low-scale *l* feature maps to make their dimensions consistent with the mesoscale *m* feature maps so that different types of spatial information can be fused; then, the nearest-neighbour interpolation is used to adjust the dimensions of the high-scale *s* feature maps to maintain consistency with the mesoscale. Finally, by splicing the three scales of feature maps in the channel dimension, a feature tensor that integrates the information of low, medium and high scales is obtained. This multi-scale feature fusion method can effectively improve the network's ability to perceive information of different dimensions and semantic levels, thus providing richer contextual information for subsequent tasks and enhancing the robustness and accuracy of the model. The module focuses on adjusting the scale through adaptive pooling and interpolation, splicing low, medium and high scale features in the channel dimension, providing rich multi-scale feature information, and focusing on simple spatial dimension alignment and feature splicing. The role of the ASF2 module is to process and fuse the feature maps from different scales. Firstly, the feature maps from different scales are adjusted to the same number of channels by 1x1 convolution, and their spatial dimensions are aligned by interpolation. Next, they are fused in the depth dimension using 3D convolution, after which they are further processed by batch normalisation and activation functions. Finally, the final fused feature map is obtained after 3D max-pooling. This module is designed to enhance the model's ability to perceive information at different spatial scales through multi-scale feature fusion, emphasising the fusion of spatial and depth dimensions to provide a deeper level of feature fusion. The difference between ASF1 and ASF2 is shown in Table 1.

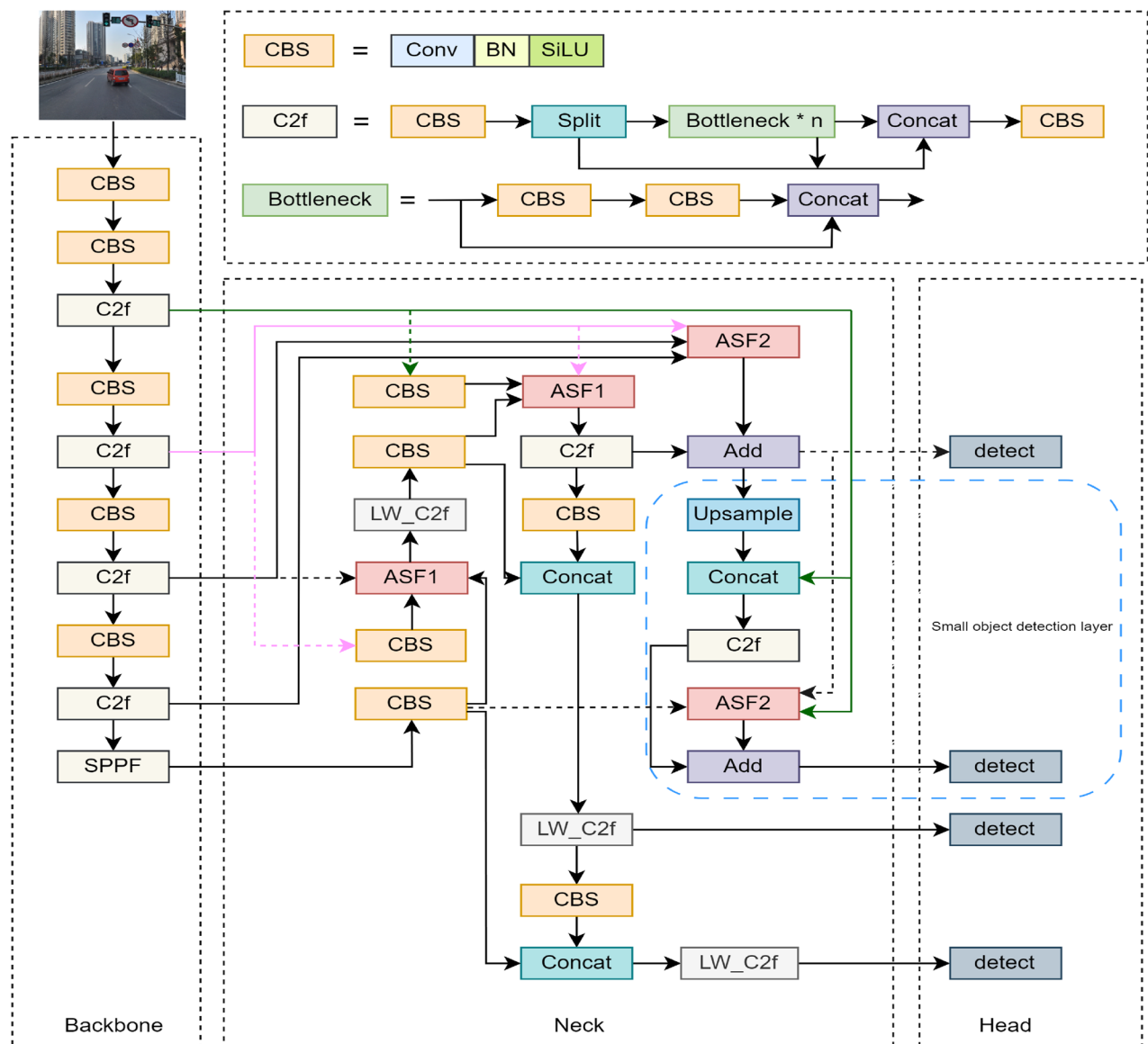


Fig. 1. Improved YOLOv8 structure diagram. By introducing the ASF module in the neck part, the network has stronger feature representation ability, and combining the ASF module with the small target detection layer can improve the focus on small target features. Using LW_C2f to replace the original C2f enables it to capture various spatial features at multiple scales and reduce the number of model parameters and model size.

Comparison dimension	ASF1	ASF2
Input	Three scales: low, medium, high	Multi-scale feature maps
Integration method	Integration of spatial features at different scales	Fusing spatial and depth dimension information
Structural complexity	Relatively simple	Relatively complex
Objective	Rich context information	Integration of spatial and depth dimensions

Table 1. The difference between ASF1 and ASF2.

AFP

The AFP module is a combination of the ASF module and the P2 small target detection layer. This article calls the combined module the AFP module. The ASF module is introduced first, and the network structure is shown in Fig. 2.

As shown in the Fig. 2, the ASF module can be divided into three parts: SSFF, TFE, and CPAM. SSFF can combine global or high-level semantic information from multiple scale images, adjust the number of channels

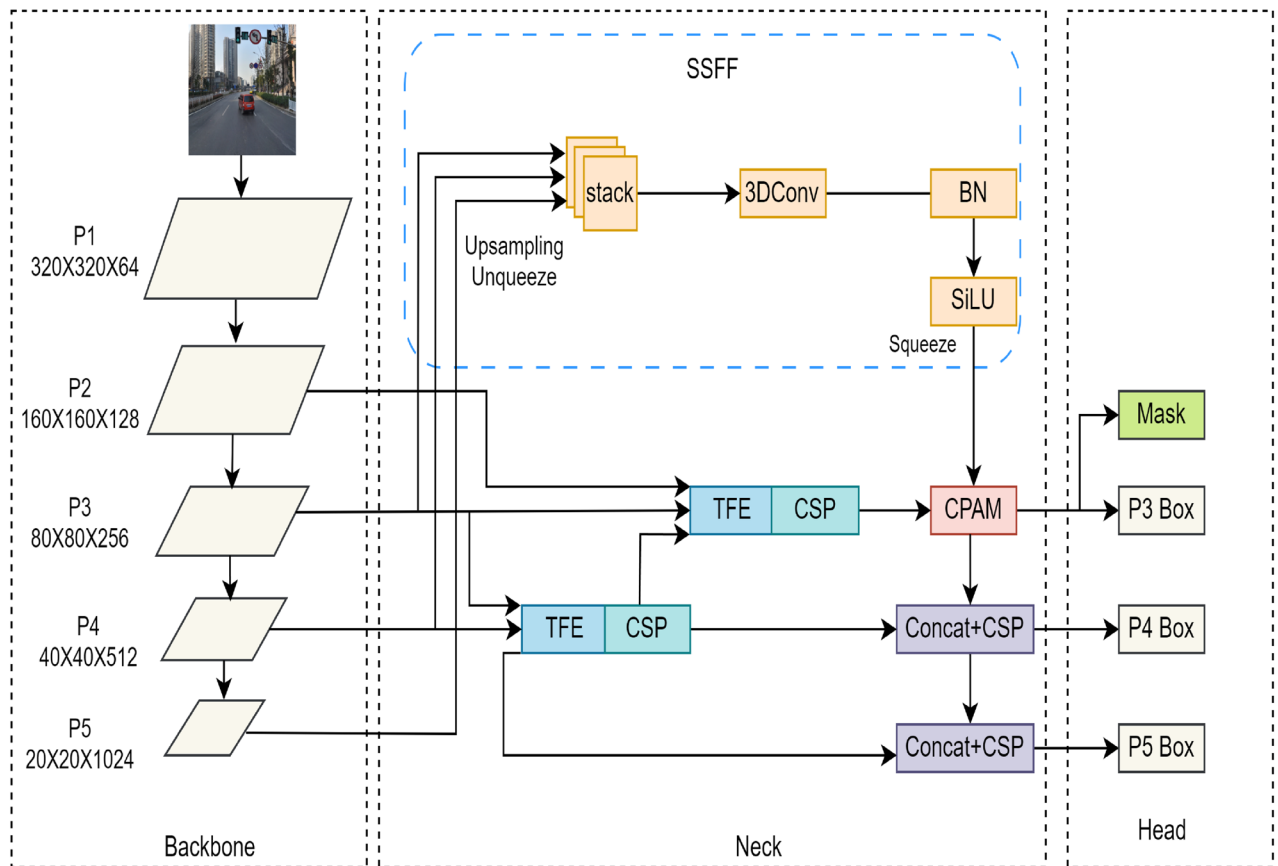


Fig. 2. ASF model structure diagram. The framework mainly consists of a Scale-Sequence Feature Fusion (SSFF) module, the Triple Feature Encoding (TFE) module, and the Channel and Position Attention Model (CPAM) based on the CSPDarkNet backbone and the YOLO head.

and sizes of the P4 and P5 feature layers through 1x1 convolution and nearest neighbor interpolation, and then use the unsqueeze method to increase the dimension of each feature layer, and use 3D convolution, 3D batch normalization, and SiLU activation function to complete scale sequence feature extraction. The scaled images as input to SSFF can be obtained by the following methods:

$$F_{\sigma}(w, h) = G_{\sigma}(w, h) \times f(w, h) \quad (1)$$

$$G_{\sigma}(w, h) = \frac{1}{2\pi\sigma^2} e^{-(w^2+h^2)/2\sigma^2} \quad (2)$$

where $f(w, h)$ represents 2D input images with width w and height h . $F_{\sigma}(w, h)$ is generated by smoothing with a series of convolutions using a 2D Gaussian filter $G_{\sigma}(w, h)$. σ is the scaling parameter of the standard deviation of the 2D Gaussian filter used for convolution.

The TFE module, splits large, medium and small features to increase the large size feature map and performs feature magnification to enhance detailed feature information. Detailed information of small targets is captured by splicing large, medium and small features of different sizes in spatial dimension to enhance the detection of dense targets. The formula for the TFE module is as follows:

$$F_{TFE} = \text{Concat}(F_l, F_m, F_s) \quad (3)$$

F_l, F_m and F_s denote large, medium, and small-size feature maps, respectively. F_{TFE} results from the concatenation of F_l, F_m and F_s . F_{TFE} has the same resolution as and three times the channel number of F_m .

The CPAM module can extract representative feature information contained in different channels, can receive the input channel attention network from TFE, and superimpose the SSFF output with the input position attention network, and then capture the information channel and the spatial positioning of related targets respectively, thereby improving its detection and segmentation accuracy.

YOLOv8 contains 3 detection heads for detecting targets of different sizes. However, after introducing ASF into the Neck part of YOLOv8, it is found that the model performs poorly when dealing with tiny targets. As the model deepens, small target features and low-order features will be lost, and small targets usually occupy fewer pixels in the image and are more likely to be ignored or misjudged. Therefore, the P2 small target detection

layer is introduced to combine with ASF to form the AFP module, which is used to supplement more feature information and enhance the detection ability of the model, especially for small targets.

The P2 small target detection layer can be responsible for the extraction of feature maps required for smaller targets, which improves the accuracy and robustness of small target detection. By performing multiple down-sampling and up-sampling operations on the input image, a series of feature maps of different scales can be generated, enabling the model to detect targets of different sizes, and the feature maps of the P2 layer are usually cascaded or fused with those of the other layers to obtain richer feature representations, enabling YOLOv8 to detect small-target objects more effectively.

As shown in Fig. 1, the feature maps added to the small target detection layer used to detect detection are fused from the feature maps of different layers. The process is as follows: step 1, it is first up-sampled and then spliced with the P2 feature layer of the backbone network to obtain more feature information. Step 2, the P2 feature layer of the backbone network and the P3, P4 feature layers of the Neck layer are fused to better combine the detailed information of the shallow feature maps with the high-dimensional information of the deep feature maps. Finally the results of step 1, step 2 are summed and processed by an additional detection header: DETECT. Performing the new small target detection layer at the end of the Neck section, followed by up-sampling, helps prevent the use of too many parameters. At the same time, multiple feature layers are fused through a series of operations to combine deep and shallow information. This further enhances the feature fusion capability of the network and improves the detection accuracy of small targets. The introduction of additional detection heads extends the detection of traffic signs, allowing the model to detect targets containing more detailed low-order information and smaller targets.

LW_C2f

The traditional convolution operation is that each channel of the input image is convolved with a convolution kernel and then the results of all the channels are summed up to get the final output. This shows that each convolution kernel performs convolution operation with all channels of the input, which is computationally intensive, especially in the case of having a large number of input channels and convolution kernels. In this paper, we propose a new convolution method, Lightweight Convolution (LWConv), which aims to ensure accuracy while reducing the number of parameters in the convolution. The specific process is as follows: Assuming that the number of channels of the input feature is C , we first perform a grouping operation to transform the shape of the input tensor from (bs, C, h, w) to (bs, g, ch, h, w) , where $ch = C / g$, bs represents the batch size, g represents the number of groups, ch represents the number of channels in each group, and h and w are the height and width of the feature map, respectively. Next, the input channel number C is evenly divided into 4 groups, and the feature maps of each group are convolved with convolution kernels of different sizes: the first group of feature maps is convolved with a 1×1 convolution kernel, the second group of feature maps is convolved with a 3×3 convolution kernel, the third group is convolved with a 5×5 convolution kernel, and the fourth group is convolved with a 7×7 convolution kernel. The application of these different convolution kernels helps capture spatial features at different scales. After the convolution operation is completed, the four groups of feature maps are stacked according to the channel dimension to generate a new feature map with a shape of $(bs, g \times ch, h, w)$. Subsequently, the shape of the new feature map is restored to (bs, C, h, w) by shape rearrangement, that is, the convolution results of all groups are merged. Finally, through a point-by-point convolution operation, the channel information is fused and the number of channels of the output tensor is restored to the number of channels C at the input, so that the shape of the output tensor is consistent with the input.

Compared with traditional convolution, LWConv introduces multiple convolutions with different kernel sizes to capture various spatial features at multiple scales, and uses larger convolution kernels to expand the receptive field, thereby enhancing its ability to model long-range dependencies. By dividing the input channels into multiple groups, LWConv enables the calculation of each group to be performed independently, thereby optimizing the calculation process. Although more convolutional layers have been introduced, the parameters of each convolution kernel are fewer, effectively reducing memory consumption and computational complexity. In addition, pointwise convolution further enhances the information exchange between channels, which facilitates feature fusion. Importantly, we divide the channels into four groups to balance parameter reduction and feature diversity. When the number of groups is too small (e.g., $g = 2$), the diversity of receptive field sizes is limited, hindering the model's ability to capture rich multi-scale features. Conversely, using too many groups (e.g., $g = 8$) reduces the number of channels per group, degrading feature representation and increasing structural complexity. Setting $g = 4$ provides an effective trade-off, offering sufficient scale diversity while maintaining a low parameter count and manageable complexity. The structure of LWConv is shown in Fig. 3.

C2f is an important part of YOLOv8, which is mainly responsible for feature extraction and downsampling. By integrating LWConv into the bottleneck of C2f, and replacing the two ordinary convolution modules in the bottleneck with LWConv, the redundancy introduced by the standard convolution in the bottleneck can be effectively reduced, and the number of parameters can be reduced under the premise of not weakening the feature extraction capability. Reduce the number of parameters. At the same time, adopting different number of channels for different scales of models can obtain richer gradient flow information. The structure of LW_C2f is shown in Fig. 4. This modification emphasizes spatial feature extraction capability and optimizes the model size without compromising detection performance.

Wise-IoU

The role of the loss function is to calculate the gap between the forward computation result and the real value of each iteration of the neural network, so as to guide the next training step in the right direction. Therefore, optimising the loss function can greatly optimise the performance of the network. The default loss function for YOLOv8 is CIoU, and the formula for CIoU is as follows:

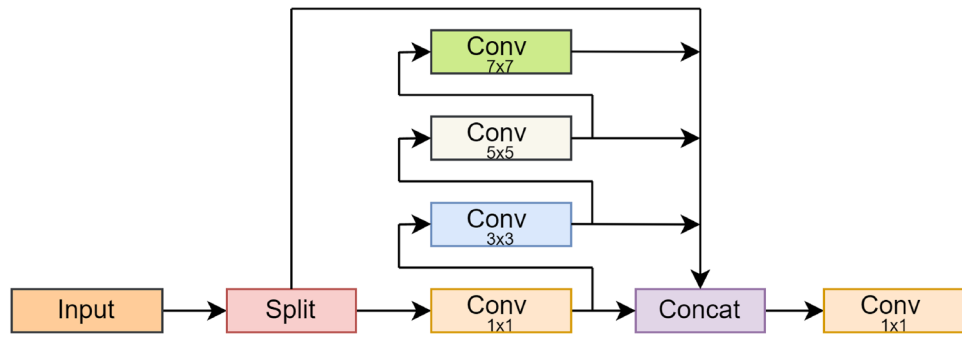


Fig. 3. LWConv structure diagram.

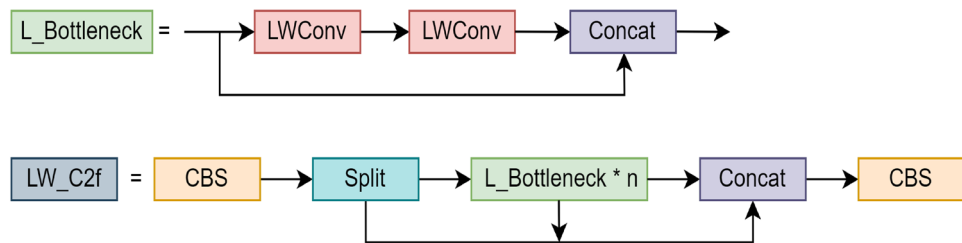


Fig. 4. Network structure diagrams for L_Bottleneck and LW_C2f.

$$L_{\text{CIoU}} = L_{\text{IoU}} + \frac{\sigma(b, b^t)}{(w^c)^2 + (h^c)^2} + av \quad (4)$$

$$L_{\text{IoU}} = 1 - \frac{|b \cap b^t|}{|b \cup b^t|} \quad (5)$$

$$v = \frac{4}{\pi^2} \left(\arctan \frac{w^t}{h^t} - \arctan \frac{w}{h} \right)^2 \quad (6)$$

$$a = \frac{v}{L_{\text{IoU}} + v} \quad (7)$$

Where w, h, b are the width, height and centroid of the boundary of the prediction frame respectively, and w^t, h^t, b^t are the width, height and centroid of the boundary of the real frame respectively. σ denotes the Euclidean distance between the centroid of the boundary of the prediction frame and the centroid of the boundary of the real frame. w^c, h^c are the width and height of the smallest outer rectangle of the prediction frame and the real frame. The CIoU loss function is more computationally complex than the traditional IoU loss function. It needs to calculate the distance and angle difference between target detection frames, which increases the computational time and resource consumption and may not be sensitive enough when dealing with small targets.

In this study, the Wise-IoU loss function is introduced. Wise-IoU can more accurately assess the degree of overlap between targets, and can give higher importance to the overlap of small targets, which reduces the possibility of omission and misdetection. The overlap between small targets is made more sensitive by applying additional weighting to the intersection region. This enhances the accuracy of small target detection, improves the detection rate of the algorithm on small targets, and can better adapt to targets of different shapes and sizes, and the additional weighting of the intersection region improves the robustness to small targets, so that the algorithm can still accurately detect small targets in complex scenarios. Wise-IoU has three versions, v1, v2, and v3. $Wise - IoU_{v1}$ constructs an attention-based bounding box loss, and $Wise - IoU_{v2}$ and $Wise - IoU_{v3}$ add a focusing mechanism by constructing a calculation method for the focusing coefficients. v2 uses a monotonically focusing mechanism, and v3 uses a dynamic non-monotonic focusing mechanism. In this study, $Wise - IoU_{v3}$ is adopted because the dynamic non-monotonic focusing mechanism can flexibly adapt the gradient assignment strategy to varying training conditions, providing a better balance between hard and easy examples, thus improving the accuracy of bounding box regression. In contrast, v1 and v2 use fixed or monotonic focusing strategies that lack adaptability to complex and noisy data distributions, limiting their effectiveness in challenging scenarios. It can make the gradient gain assignment strategy that best fits the current situation at every moment, effectively preventing the low-quality examples from generating large and harmful gradients. The formula for $Wise - IoU_{v3}$ is as follows:

$$L_{Wise-IoU_{V1}} = L_{IoU} \exp \frac{(x - x_{gt})^2 - (y - y_{gt})^2}{(W_g^2 + H_g^2)} \quad (8)$$

$$L_{Wise-IoU_{V3}} = r L_{Wise-IoU_{V1}} \quad (9)$$

$$\beta = \frac{L_{IoU}^*}{L_{IoU}} \quad (10)$$

$$r = \frac{\beta}{\delta \alpha^{\beta-\delta}} \quad (11)$$

where (x_{gt}, y_{gt}) denote the position of the target frame, W_g, H_g are the size of the minimum closed frame, and the non-monotonic focusing coefficients β and $L_{Wise-IoU_{V1}}$ are utilized to construct $Wise - IoU_{v3}$.

Experiments

Dataset

TT100K dataset. TT100K is a traffic sign dataset in China, which is compiled and published by Tsinghua University in cooperation with Tencent, and it provides 100000 pictures containing 30000 traffic signs with the resolution of 2048×2048, which are rich in shooting scenarios, with varying lighting conditions and weather variations, and with a complex shooting background containing a large amount of interfering information. In this paper, 9457 pictures with 42 types of traffic signs are selected, of which 6598 pictures are used for training, 1889 pictures are used for validation, and 970 pictures are used for testing.

Evaluation indicators

This study uses precision, recall, multi-category average precision (mAP), model parameter count (M) and model size (MB) as evaluation indicators. The specific calculation method is as follows:

$$Precision = \frac{TP}{TP + FP} \quad (12)$$

$$Recall = \frac{TP}{TP + FN} \quad (13)$$

$$AP = \int_0^1 Precision(t) dt \quad (14)$$

$$mAP = \frac{\sum_{n=1}^N AP_n}{N} \quad (15)$$

TP indicates the number of samples that are positive and predicted to be positive, FP the number of samples that are negative but predicted to be positive, and FN the number of samples that are actually positive but predicted to be negative. AP (average precision) is the average precision of object detection. It synthesizes the performance of the model under different precision and recall conditions, reflecting the balance between accuracy and recall.

Experimental settings

The experimental environment of this study is shown in Table 2 below, the hardware specifications are Intel Platinum 8255C @ 2.50GHz and RTX 3080 (10GB), and the memory capacity is 30GB. in terms of software, ubuntu is used as the operating system, the programming language is pytorch, and the framework is PyTorch2.0.0.

In this study, the hyperparameter settings in the model training process are shown in Table 3, the optimiser is SGD, the input image size is 640*640, epoch is 300, the momentum parameter is 0.937, the initial learning rate is 0.01, and the batch size is 8.

Experimental configuration	Name	Information
Hardware	CPU	Intel Platinum 8255C @ 2.50GHz
	GPU	RTX 3080
	VRAM	10 GB
	Memory	30 GB
Software	Operating system	ubuntu20.04
	Python Version	3.8
	PyTorch Version	2.0.0
	CUDA Version	11.8

Table 2. Experimental configuration.

Training parameter	Parameter value
Optimiser	SGD
Imgs	640
Epochs	300
Batchsize	8
Momentum	0.937
Initial learning rate	0.01

Table 3. Training parameter setting.

YOLOv8n	AFP	LW_C2f	Wise-IOU	Precision(%)	Recall(%)	mAP50(%)	M(params)	Size(MB)
✓				77.4	69.9	77.1	3.1	6.4
✓	✓			81.9	72.6	81.3	2.7	5.9
✓		✓		76.1	70.4	78.6	2.8	6.0
✓			✓	77.6	70.5	77.7	3.1	6.2
✓	✓	✓		82.2	74.7	82.1	2.6	5.6
✓	✓	✓	✓	82.6	76.1	82.8	2.5	5.6

Table 4. Ablation experiment results.

Experimental results and analysis

Ablation experiment In order to verify the impact of the improvements proposed in this paper on the performance of traffic sign detection, single or multiple improved ablation experiments were added to the TT100K dataset as shown in Table 4.

By introducing the AFP module, the model’s capability in detecting small objects is significantly enhanced. The integration of the ASF structure with the P2 detection layer effectively strengthens the Neck’s ability to represent multi-scale features of small targets. By emphasizing critical features and suppressing irrelevant ones, this module enables accurate localization even in complex backgrounds or when target sizes are small. As shown in Table 4, the introduction of AFP leads to increases of 4.5%, 2.7%, and 4.2% in Precision, Recall, and mAP50, respectively, while reducing the number of parameters and model size by 0.4M and 0.5MB.

In addition, the LW_C2f module combines the LWConv and C2f structures. By introducing multi-scale convolution kernels , it improves the model’s spatial perception and receptive field, thereby enhancing its ability to model long-range dependencies. LWConv employs a grouped convolution strategy with fewer parameters per group, effectively reducing redundant computation. Furthermore, pointwise convolutions are used to promote information exchange across channels. This module ultimately reduces parameter count and model size by 0.3M and 0.4MB, respectively, and despite a slight decrease in Precision, it results in improvements of 0.5% and 1.5% in Recall and mAP50, demonstrating a favorable balance between efficiency and performance.

Furthermore, Wise-IoU is adopted as the bounding box regression loss function. It introduces a dynamic non-monotonic focusing mechanism to enhance localization accuracy by dynamically allocating gradient gains based on the outlier degree of anchor boxes. Low and high-quality anchor boxes receive smaller gradients to prevent misleading optimization, while the focus is placed on medium-quality samples to suppress harmful gradients and improve regression stability. With the integration of Wise-IoU, Precision, Recall, and mAP50 increase by 0.2%, 0.6%, and 0.6%, respectively, along with a 0.2MB reduction in model size, confirming its effectiveness in optimizing localization.

Overall, the AFP module enhances small object detection capability, the LW_C2f module achieves efficient multi-scale representation, and Wise-IoUv3 optimizes the localization process. The synergy of these three components leads to improvements of 5.2%, 6.2%, and 5.7% in Precision, Recall, and mAP50, respectively, while reducing the parameter count and model size by 0.6M and 0.8MB. This indicates that the improvement strategy proposed in this paper not only achieves significant improvement in detection accuracy, but also effectively reduces the number of parameters and model size, showing better comprehensive performance.

Models Comparison In order to further verify the effectiveness of the improved model, this paper compares the improved model with other mainstream models for target detection such as Faster-RCNN, Deformable DETR, YOLOv3, YOLOv5s, and YOLOv7 on the TT100K dataset using the same experimental equipment, experimental parameters, and training strategy, and the experimental results are shown in Table 5.

From Table 5, it can be seen that on the TT100K dataset, YOLOv8n outperforms Faster-RCNN, Deformable DETR, YOLOv3, and YOLOv7 in terms of the number of parameters, model size, Precison, Recall, and mAP50. Although YOLOv5s improves 4.1% and 4.3% in Precison and mAP50, respectively, compared to YOLOv8n, it is inferior to YOLOv8n in the number of parameters and model size, while this paper’s improved model not only outperforms YOLOv8n in all aspects but also outperforms YOLOv5s, and 3.5%, it is not as good as YOLOv8n in terms of the number of parameters and model size, while the improved model in this paper is not only better than YOLOv8n in all aspects, but also better than YOLOv5s, with the model size and the number of parameters

Methods	Precision (%)	Recall (%)	mAP50 (%)	M (params)	Size (MB)
Faster-RCNN	69.3	63.6	69.4	27.6	108.6
Deformable DETR	73.2	69.5	76.3	20.6	41.7
YOLOv3	74.9	68.3	75.5	8.9	18.7
YOLOv5s	81.5	75.6	81.4	7.13	14.7
YOLOv7	73.1	68.6	73.6	37.4	75.2
YOLOv8n	77.4	69.9	77.1	3.1	6.4
Ours	82.6	76.1	82.8	2.5	5.6

Table 5. Comparative experiment results.



Fig. 5. TT100K dataset detection results (top row YOLOv8n model, bottom row improved model).

only 1/3 of that of YOLOv5s, and the Precision and Recall and mAP50 are tuned up by 1.1%, 0.5% and 1.4%, respectively, compared with YOLOv5s.

The experiments show that the improved model in this paper can accurately improve the detection accuracy of traffic signs while significantly reducing the number of model parameters, and has more excellent performance for the precise location of small targets and classification of small-scale traffic targets.

Visual analysis In order to visualise the effect of the improvement in this paper, some pictures are selected for detection on the TT100K dataset, and the selected pictures include single target, multi-targets, and small targets at a distance, and the improved model is compared with the original model, and the detection results are shown in Fig. 5. From the detection results in Fig. 5, it can be clearly seen that the improved model has achieved significant improvement in several aspects compared with the original model. Firstly, the traffic sign in the first picture is pl40, which is incorrectly detected as pl60 by the benchmark model. This misdetection problem may be due to the limitations of the original model in dealing with the scale and shape of traffic signs, which leads to the inability to effectively distinguish between similar signs. In contrast, the improved model in this paper introduces a multi-scale feature fusion method to obtain richer contextual information, which can correctly



Fig. 6. Heat map (from left to right, original image, YOLOv8n model, improved model).

identify p140 traffic signs, and the accuracy of detection is effectively improved. Next, the last two images demonstrate the leakage problem of the original model in the detection process. Specifically, the original model fails to detect the P11 traffic sign in the second picture, and the original model fails to detect the Pn traffic sign in the third picture. After analysis, the main reason for the missed detection can be attributed to the insufficient ability of the original model in dealing with small targets, especially when the targets are small or partially obscured, the original model is difficult to accurately detect these signs, and the robustness of the model is poor. In contrast, the improved model in this paper adds the AFP module, which not only better captures the detail information of small targets, but also enhances the saliency of small target features when the target is small or partially occluded, which makes the improved model better able to extract the information of small-target traffic signs, and avoids the phenomenon of missed detection. In the last two pictures in Fig. 5, the improved model is able to accurately detect two traffic signs, P11 and Pn, and the accuracy in traffic sign detection is also improved.

The comparison results in Fig. 5 show that the improved model has made significant progress in solving the problem of misdetection and omission of small targets that existed in the original model. By introducing the AFP module, the multi-scale feature fusion method can not only better capture the detail information of small targets, but also enhance the saliency of small target features when the target is small or partially occluded. Meanwhile, the Wise-IOU loss function is introduced to make the model more sensitive to the overlap between small targets, which improves the robustness of the model to small targets, enabling the model to identify small targets more accurately during the training process, and effectively distinguish and detect individual targets even when there is a high degree of overlap between targets.

Heatmap analysis The Grad-CAM heat map can help to understand how much the model pays attention to different regions during the inference process, which can help to further understand the model's attention mechanism and decision-making basis.

In view of this, the Grad-CAM heat maps of the two models before and after the improvement were compared, and the results are shown in Fig. 6. In the first row of the comparison graph in Fig. 6, it can be seen that both models before and after the improvement have good attention to traffic signs, but the benchmark model YOLOv8n has some distraction in its attention, which focuses on some irrelevant factors such as cars on the road in addition to traffic signs, this is due to the fact that the categorical labels of the benchmark model contain cars, traffic signs and so on, whereas the attention of the improved model is more focused, with the labels are focused on traffic signs, and in the figure it is represented as focusing on 3 traffic signs. In the second row of the comparison graph, it can be seen that the benchmark model focuses on pedestrians and cars on the road in addition to traffic signs, whereas the improved model focuses on traffic signs, and also focuses on the tiny traffic signs behind them, which suggests that the improved model is more accurate in focusing on the key targets and can accurately identify traffic signs for smaller traffic signs as well.

Conclusion

To address the limitations of YOLOv8 in traffic sign detection, including weak perception of small objects, redundant parameters, and the loss function's excessive sensitivity to low-quality samples, this paper proposes three improvement strategies and develops a lightweight and high-precision small object detection model. First, AFP module is designed by integrating the attention-scale fusion structure from ASF-YOLO with the P2 layer of YOLOv8, thereby enhancing the Neck's ability to represent multi-scale features of small objects. This module significantly improves the model's detection performance for small-scale traffic signs in complex backgrounds while maintaining overall inference efficiency. Second, a lightweight convolutional structure, LW_C2f, is designed, which employs multi-scale convolutional kernels and a group convolution mechanism to reduce redundant computation and model size, while preserving critical semantic features and improving discrimination in small object regions. Finally, a novel regression loss function, Wise-IoU, is adopted. By introducing a dynamic non-monotonic focusing mechanism, it suppresses the negative gradient interference from low-quality anchors and emphasizes medium-quality bounding boxes, thereby enhancing the stability and robustness of the regression process. Experimental results demonstrate that, compared to the original model, the proposed approach achieves improvements of 5.2%, 6.2%, and 5.7% in Precision, Recall, and mAP50, respectively, on the TT100K dataset. In addition, the number of parameters and the model size are reduced by 0.6M and 0.8MB, respectively, confirming the effectiveness of the proposed method in enhancing detection performance while simultaneously reducing model complexity.

Despite the strong detection performance of the proposed model on the TT100K dataset, particularly in its improved ability to detect small traffic signs in complex backgrounds, its robustness under adverse weather conditions remains limited. In scenarios such as rainy or foggy environments, image quality often deteriorates due to blurring, occlusion, and reduced contrast, which can compromise the saliency of target objects and negatively impact detection accuracy and stability. Future research may explore the integration of image enhancement techniques with deraining or dehazing capabilities to improve the model's generalization and reliability in challenging weather conditions.

Data availability

The datasets analyzed during the current study are available in TT100K2021(<https://cg.cs.tsinghua.edu.cn/traffic-sign/>).

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Author contributions

G.H.W and P.J. designed the experimental methods. P.J. wrote the main manuscript text. Z.W.Q. conducted the experiments and X.H.L. was responsible for the statistical analyses and data compilation. All authors were involved in writing and reviewing the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to G.W.

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